RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Department of Mechanical Engineering

Date: 02/04/2025	**CIE – Test 1**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part – A			
1	According to which principle the electrons fill the lowest energy orbital first before moving to higher energy ones.	1	1	1
2	In _____ material the conduction and valence bands overlap so electrons can easily pass into the conduction band.	1	1	1
3	Define unit cell.	1	1	1
4	Write the classification of composites based on matrix.	2	1	1
5	The Burgers Vector is parallel to the dislocation line, the type of line defect is _____.	1	1	1
6	If an atom is missing from its normal site in the matrix, the defect is called _____ defect.	1	1	2
7	Define thermal conductivity with equation.	2	2	2
8	In dielectric materials the movement of electrons in response to an external electric field is _____ type of polarization mechanism.	1	1	2
	Part – B			
1(a)	With the help of neat sketches explain the Primary Interatomic bonds with examples.	10	2	1
2(a)	Classify solid materials explain them briefly with examples.	10	2	1
3(a)	Calculate Atomic packing factor for FCC unit cell.	5	3	2
3(b)	Classify and explain different crystal imperfections.	5	2	2
4(a)	Compare the properties and applications of metals and polymers.	10	3	1
5(a)	Illustrate the following thermoelectric effects: i) Seebeck effect ii) Peltier effect.	5	3	2
5(b)	Discuss different types of Insulating materials highlighting their applications.	5	2	2

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Department of Biotechnology

Date: 2nd April 2025	**Closed Book Offline Quiz & Test – 1**	**Max. Marks: 60**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Biosafety Standards and Ethics (Basket Course)		**Course Code: BT242TC**

S No	Questions	M	BT	CO
	Quiz			
1	What is the role of the Institutional Biosafety Committee (IBC) in India?	1	2	2
2	The Recombinant DNA Advisory Committee (RCGM) is a part of which Department of Govt of India?	1	1	2
3	The committee of an Institutional Biosafety Committee (IBC) comprise of _____ and _____.	1	1	2
4	Biosafety levels indicate _____.	1	1	1
5	_____ is the primary mode of transmission of many viral biohazards.	1	1	1
6	Which PPE is essential for BSL 4 labs?	1	2	1
7	In class II biosafety cabinet, which direction does the air flow?	1	2	2
8	The Cartagena Protocol was adopted in which year?	1	1	2
9	Mention the 3 types of mechanism which takes place in HEPA filters.	2	3	2
	Test			
1	Comment on various types of Biohazards, and their impact on human health.	10	2	1
2	Explain Biosafety level II class B2 with a neat diagram.	10	2	1
3	Discuss the importance of biosafety guidelines in India and how they contribute to public health and safety.	10	3	2
4	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	10	3	2
5	How do the Biosafety guidelines of the Government of India align with international biosafety standards? Discuss any significant differences or similarities.	10	2	2

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Department of Civil Engineering

Date: 02/04/2025	**CIE – I (Quiz + Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part – A			
1	List the components of environment.	2	1	1
2	Define biodiversity.	2	2	1
3	Differentiate between primary and secondary pollutants.	2	3	2
4	The permissible day and night noise levels for a silent zone in India are _____ dB and _____ dB respectively.	2	2	2
5	List any four sources of solid waste.	2	2	1
	Part – B			
1	Classify different types of eco system and explain any two briefly.	10	2	1
2	Define solid waste management and list the objectives of solid waste management.	10	2	2
3	Briefly explain the following safety terminologies. (i) Accident (ii) Ergonomics (iii) First aid (iv) Hazard (v) Safety inspection	10	2	2
4	Define noise pollution. List the effects of noise pollution and write the control measures.	10	2	2
5	With a neat sketch explain carbon cycle.	10	2	1

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Department of Biotechnology

Date: 05th July 2023	**CIE – 1**	**Max. Marks: 25**
Semester: IV	**UG**	**Duration: 60 Min**
Course Title: Environmental Technology		**Course Code: 21BT42A**

S No	Questions	M	BT	CO
1	State the SDG's promulgated by United Nations.	5	2	1
2	Discuss in brief the different components which constitute the environment.	5	3	1
3	State the significance of environmental education and NGO's in conserving the environment.	5	2	1
4	Explain the step by step process of Environmental Impact Assessment with flow chart.	5	2	1
5	Discuss the objectives and applications of ISO 14000 series in Environmental Management System (EMS).	5	2	1